1 Results

1.1 Gathered data

1. Structure of data from Silvics
   (a) Angiosperm (124 total ; 79% (98/124) with masting data; Masting and non-masting split almost equally (52%(51/98)); Gymnosperm (64 total; 95% (61/64) with masting data; almost all species are masting (89%(54/61))
2. Number of traits for different hypotheses
   (a) For angiosperm, we have data for:
      i. Predation satiation: seed weight, fruit size, oil content, seed dispersal mode, seed dormancy
      ii. Pollination coupling: pollination mode, reproductive type
      iii. Resource matching: drought tolerance, leaf longevity
   (b) For gymnosperm, we have the same data as for angiosperm, but we only have oil content for masting species and all the gymnosperm species are wind pollinated.

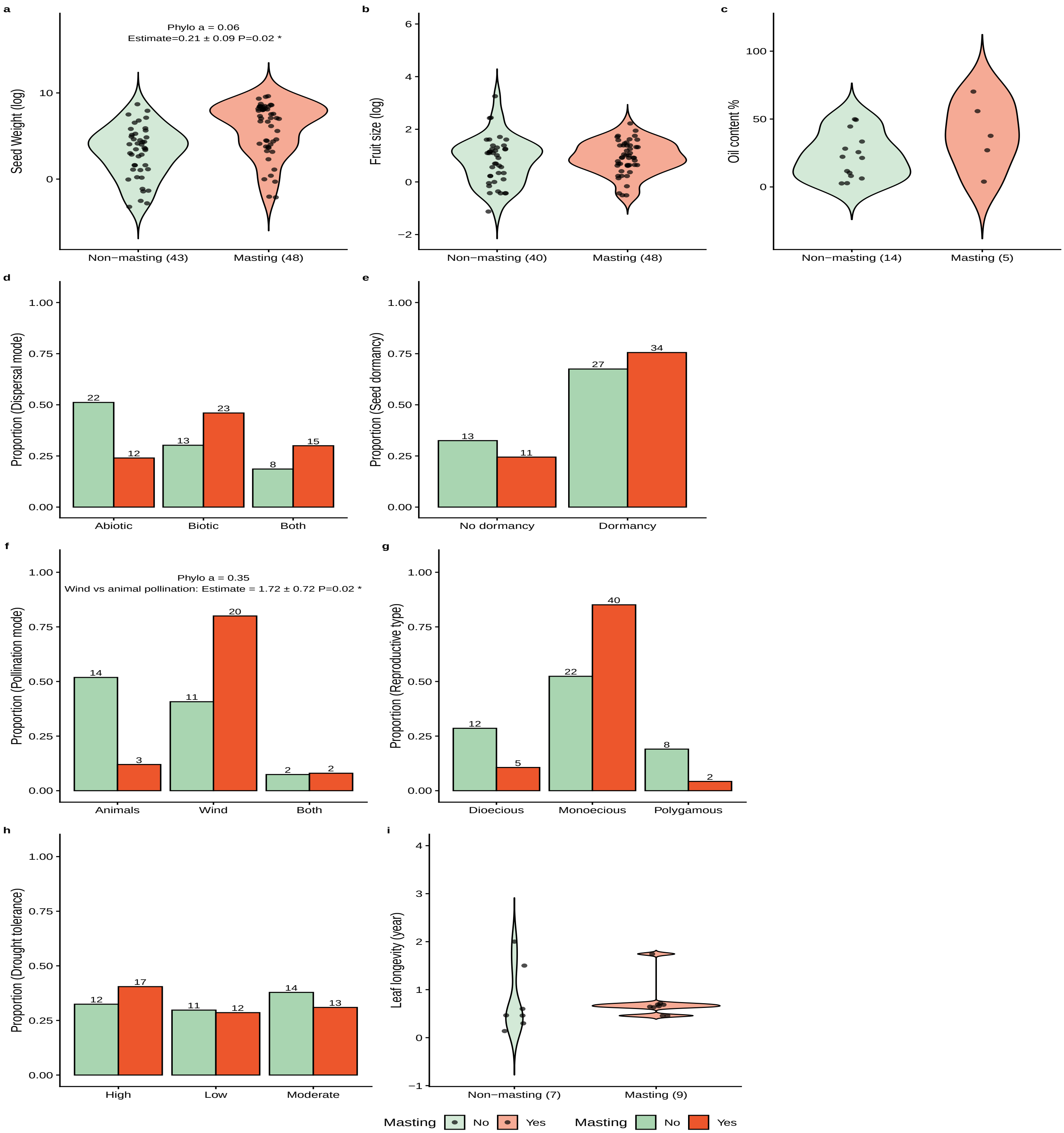

Figure 1: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf longevity and masting based on phylogenetic logistic regression for angiosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms, the y-axis of the histograms represents the proportion of each category within either the non-mast or mast group. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter α, and associated P-value for traits with a P-value < 0.5. (a)-(e) are traits related to predation satiation hypothesis; (f), (g) are traits related to pollination coupling hypothesis; (h), (i) are traits related to resource matching hypothesis. Legends are shared across panels.

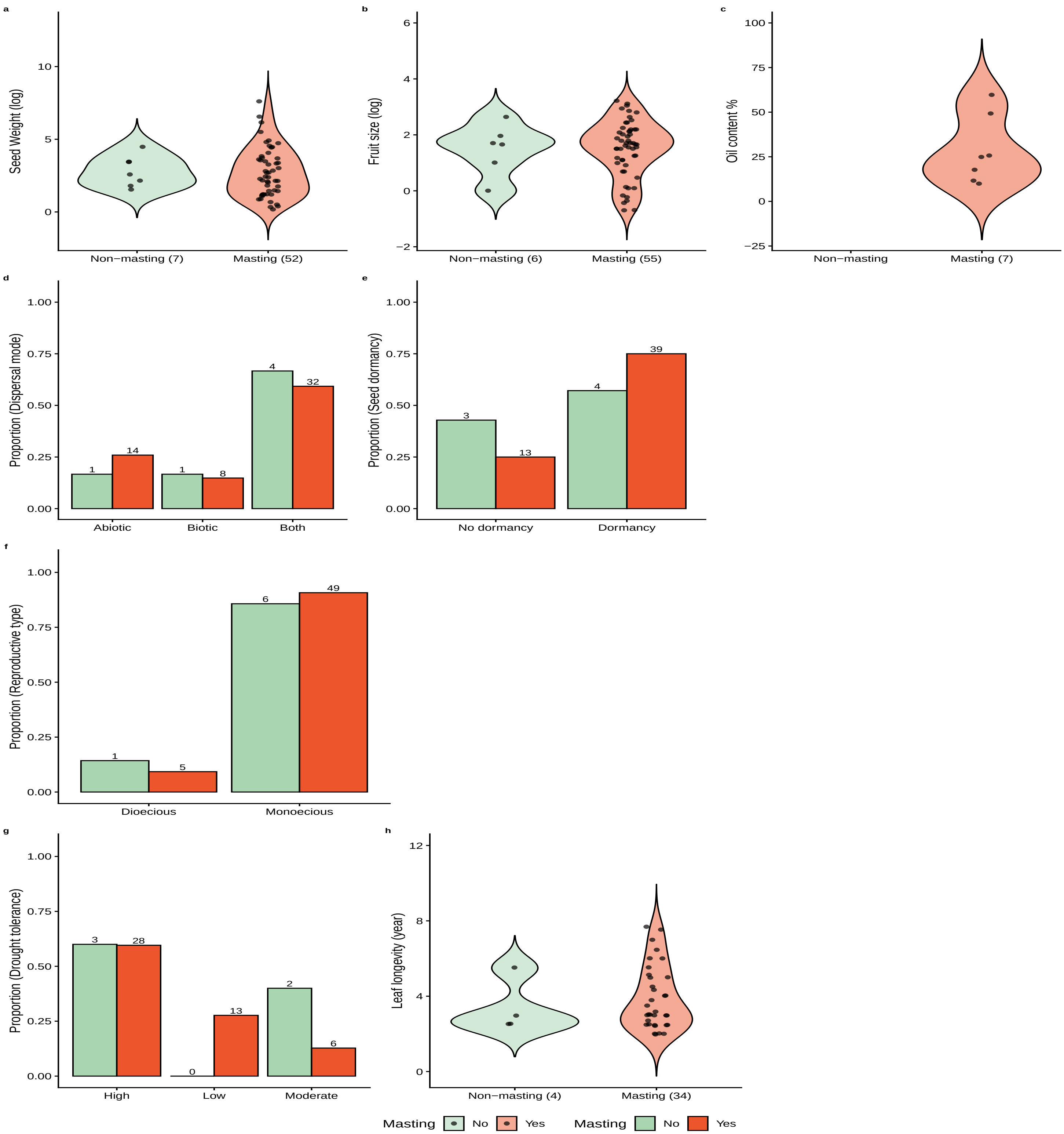

Figure 2: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Reproductive mode; (g) Drought tolerance; (h) Leaf longevity and masting based on phylogenetic logistic regression for gymnosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms, the y-axis of the histograms represents the proportion of each category within either the non-mast or mast group. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter α, and associated P-value for traits with a P-value < 0.5. (a)-(e) are traits related to predation satiation hypothesis; (f) is trait related to pollination coupling hypothesis; (g), (h) are traits related to resource matching hypothesis. Legends are shared across panels.

Trait	Predictor	Type	Estimate	SE	P	Phylo α	Half-time	N
Seed weight (log)	logSeedWeight*	Numeric	0.21	0.09	0.02	8.54	0.08	91
Fruit size (log)	logFruit	Numeric	0.08	0.30	0.80	23.25	0.03	88
Oil content (%)	Oil content (%)	Numeric	0.04	0.03	0.19	19.28	0.04	19
Dispersal mode	Biotic vs abiotic	Categorical	0.55	0.53	0.30	8.00	0.09	93
Dispersal mode	Both vs abiotic	Categorical	0.55	0.57	0.33	8.00	0.09	93
Seed dormancy	Dormant vs non-dormant	Categorical	0.03	0.50	0.95	13.07	0.05	85
Pollination mode	Wind vs animal pollination*	Categorical	1.72	0.72	0.02	47.03	0.01	52
Pollination mode	Wind vs animal pollination and animals	Categorical	1.23	1.15	0.29	47.03	0.01	52
Reproductive type	Monoecious vs dioecious	Categorical	0.91	0.68	0.18	15.00	0.05	89
Reproductive type	Monoecious vs polygamous	Categorical	-0.12	0.96	0.90	15.00	0.05	89
Drought tolerance	Low vs high drought tolerance	0.14	0.47	0.76	12.57	0.06	79	
Drought tolerance	Moderate vs high drought tolerance	-0.13	0.46	0.78	12.57	0.06	79	
Leaf longevity (years)	Leaf longevity (years)	Numeric	0.26	0.95	0.78	0.90	0.77	16

Table 1: Phylogenetic generalized linear model results for angiosperm. The structure of the model is Masting$\sim$Trait, with masting (Y/N) being the response variable and each trait being the explanatory variable seperately. Predictor for categorical trait is the difference between two levels for a specific trait. Estimate represents the log-odds associated with each predictor. SE is the standard error, and P indicates statistical significance, with $P < 0.05$ considered significant and marked with an asterisk (*) in the table. "Phylo α" represents the phylogenetic signal in the residuals estimated for each trait using phyloglm, we calculated half-time ($\log(2)/\alpha$) to compare with the tree height, which represents the time required for the influence of ancestral states to decrease by half. The tree we used is ultrametric with a standardized height of 1. If half-time is greater than tree height, it indicates strong phylogenetic signal in the residuals; if half-time is smaller than tree height, it indicates relatively weak phylogenetic signal in the residuals. N indicates the number of observations available for each trait.

Trait	Predictor	Type	Estimate	SE	P	Phylo α	Half-time	N
Seed weight (log)	logSeedWeight	Numeric	-0.03	0.25	0.91	54.43	0.01	59
Fruit size (log)	logFruit	Numeric	0.03	0.36	0.94	54.45	0.01	60
Seed dispersal	Biotic vs abiotic	Categorical	-0.83	0.81	0.30	1.93	0.36	60
Seed dispersal	Both vs abiotic	Categorical	-1.11	0.67	0.10	1.93	0.36	60
Seed dormancy	Dormant vs non-dormant	Categorical	0.71	0.70	0.32	52.15	0.01	59
Reproductive type	Monoecious vs dioecious	Categorical	-1.14	1.02	0.26	0.93	0.74	61
Drought tolerance	Low vs high drought tolerance	Categorical	11.50	94.75	0.90	0.02	36.48	52
Drought tolerance	Moderate vs high drought tolerance	Categorical	-0.25	0.58	0.67	0.02	36.48	52
Leaf longevity (years)	Leaf longevity (years)	Numeric	0.15	0.34	0.67	54.44	0.01	38

Table 2: Phylogenetic generalized linear model results for gymnosperm. The structure of the model is Masting$\sim$Trait, with masting (Y/N) being the response variable and each trait being the explainatory variable seperately. Estimate represents the log-odds associated with each predictor. SE is the standard error, and P indicates statistical significance, with $P < 0.05$ considered significant and marked with an asterisk (*) in the table. "Phylo α" represents the phylogenetic signal in the residuals estimated for each trait using phyloglm, we calculated half-time ($\log(2)/\alpha$) to compare with the tree height, which represents the time required for the influence of ancestral states to decrease by half. The tree we used is ultrametric with a standardized height of 1. If half-time is greater than tree height, it indicates strong phylogenetic signal in the residuals; if half-time is smaller than tree height, it indicates relatively weak phylogenetic signal in the residuals. N indicates the number of observations available for each trait.

1.2 Results for each hypothesis

1. Predation satiation: seed weight only with a small effect and only for angiosperm

   (a) Only seed weight is related to masting with having bigger seeds will increase the chance of being a masting species Figure (1). An increase in seed weight in the log-scale by about 3 would increase the predicted probability of masting from 0.43 to 0.6. Only within angiosperm we found this significant effect (p-value = 0.02) of seed weight.

2. Pollination satiation: wind pollination for angiosperm

   (a) we found wind pollination is related to masting for angiosperm species, wind pollinated species have a significantly higher probability (59%) of being strong masting species. Gymnosperm are all wind pollinated, and gymnosperm has a greater portion of masting species (89% (54/61)).

3. Resource matching: nothing significant

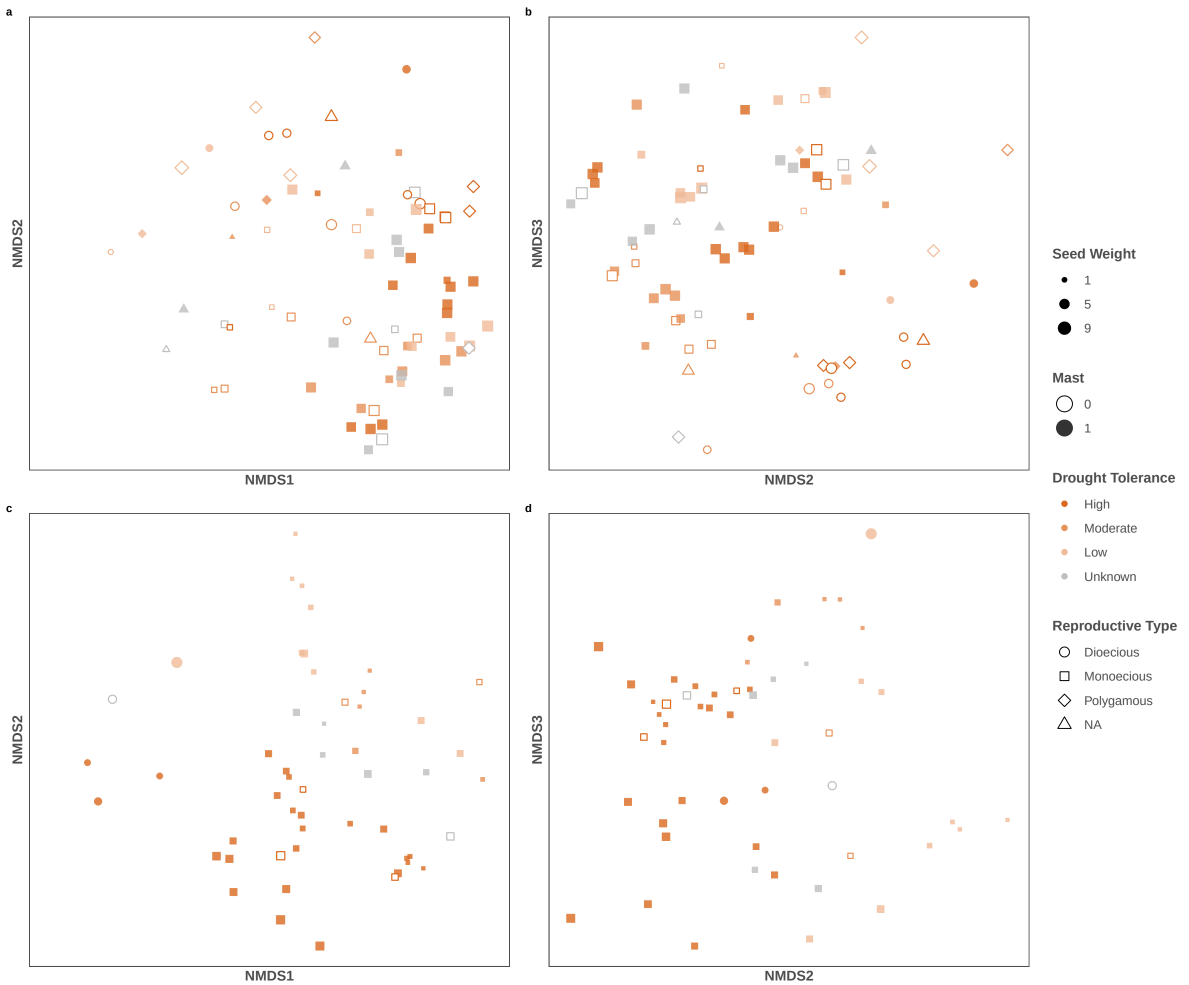

Figure 3: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and gymnosperm species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) gymnosperm NMDS1–NMDS2, and (d) gymnosperm NMDS2–NMDS3.

1.3 Multidimension-NMDS

1. No single trait drives most of the variation
   (a) We don't see masting aligned well along any axis in the NMDS analysis.
   (b) We don't have evidence that the combination of traits for any of the hypotheses is driving the pattern of masting.

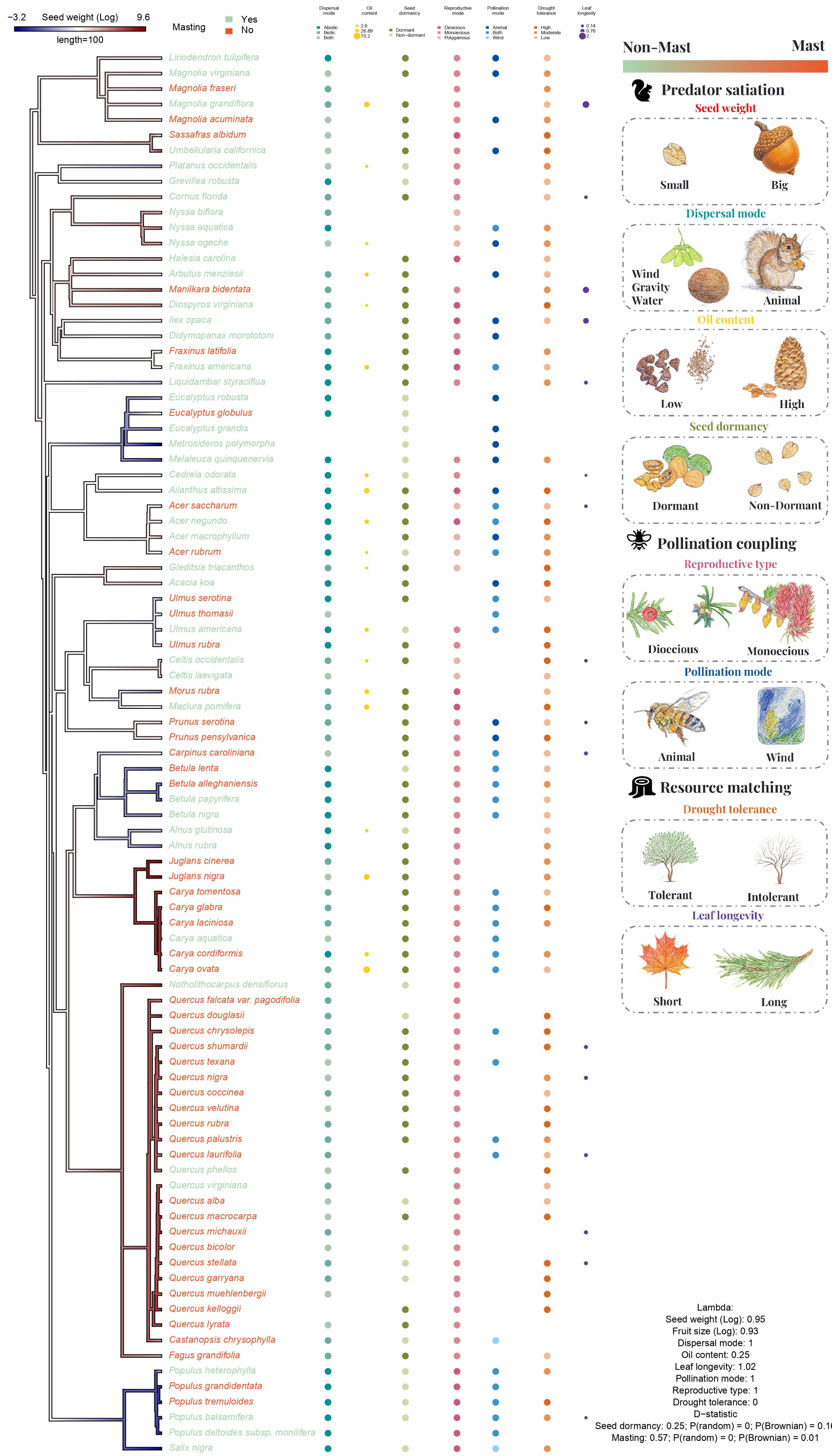

Figure 4: Phylogenetic tree for all angioperm species included in the analysis. The color scheme of the branch is the trait value for seed weight (log). The color tip is colored by masting or not. The other traits are labeled after the tip labels with different color and size. We also show the predictions of relationship between traits and masting. We calculated the Pagel's λ for continuous traits as well as multilevel categorical traits, and calculated D-statistic for binary traits

Group	Trait	(λ)
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.00
Angiosperm	Dispersal mode	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	1.00
Angiosperm	Drought tolerance	0.00
Gymnosperm	Seed weight (Log)	0.97
Gymnosperm	Fruit size (Log)	0.90
Gymnosperm	Oil content	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	1.00
Gymnosperm	Drought tolerance	1.00

Table 3: Phylogenetic Signal (Pagel's λ) for continous traits and multi-level categorical traits. $\lambda = 0$ indicates no phylogenetic signal, trait districuted independent of the phylogeny; $0 < \lambda < 1$ means moderate phylogenetic signal, trait is partially structured by the phylogeny; $\lambda = 1$ indicates strong phylogenetic signal, trait follows Brownian motion.

Group	Trait	Estimated D	P(random)	P(Brownian)
Angiosperm	Seed dormancy	0.25	0	0.16
Angiosperm	Masting	0.58	0	0.01
Gymnosperm	Masting	0.51	0.02	0.10
Gymnosperm	Seed dormancy	0.98	0.41	0
Gymnosperm	Reproductive type	-0.52	0	0.89

Table 4: Phylogenetic Signal (D-statistic) for the phylogenetic signal of binary traits, with probabilities calculated under a random phylogenetic structure and under a Brownian motion phylogenetic structure. D = 1 indicates the trait is randomly distributed across the phylogeny; D = 0 indicates the trait follows a Brownian motion model; $0 < D < 1$ indivates moderate phylogenetic signal; $D < 0$ indicates the trait is more clumped than expected under Brownian motion; and $D > 1$ indicates overdispersion. P random is the p-value for the null hypothesis that the trait is randomly distributed across the phylogeny (no phylogenetic signal), and P Brownian is the p-value for the null hypothesis that the trait follows a Brownian motion model (strong phylogenetic signal). A low p-value indicates rejection of the corresponding null hypothesis.

1.4 Phylogenetic patterns

1. Pagel's λ for continuous traits and multilevel categorical traits in Table 3, D-statistic for binary traits in Table 4.

   (a) Masting shows a moderate phylogenetic signal in both gymnosperms (estimated D = 0.51) and angiosperms (estimated D = 0.58) (Table 4).

   (b) Seed weight has strong phylogenetic signal for both angiosperm ($\lambda = 0.95$) and gymnosperm ($\lambda = 0.97$). We can see the similarity in certain group in Figure 4 it's relatively reliable estimate. Dispersal mode, leaf longevity, pollination mode and reproductive type also showed high phylogenetic signal in Table 3, but be cautious that λ for categorical traits might not be accurate.

   (c) Be cautious that λ might not be comparable among traits, not only because the way λ is calculated for continuous traits and categorical traits are different, but also because λ might depend on the data structure when sampling is uneven, take leaf longevity as an example in Figure 4,leaf longevity data is sparse and nested in *Quercus*. (5/16).